LESSON PLAN

AFCS (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Sensor units for guidance systems computer units of guidance system, reference units of guidance system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Sensor units for guidance systems computer units of guidance system, reference units of guidance system.
5. Specific Objectives	22.1.1.1Trainee will be able to understand the basic of guided weapon systems and types of targets.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of sensor block diagrams, safety: Do not touch live electronics, instructor supervision: Monitor student engagement.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a guidance system?
(b) Name two common types of guided weapon targets.
(c) How does a sensor unit differ from a reference unit?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principles of guided weapon systems • Classification of target types for guidance • Role of sensor units in target acquisition • Function of guidance system computer units • Integration of sensor and computer data	L & Ds	PPT, IB	15 min
(b)	• Reference units for inertial navigation • Reference units for celestial navigation • Reference units for terrain following • Accuracy requirements for reference data • Error sources in reference unit systems	L & Ds	PPT, PDF	20 min
(c)	• Signal processing in sensor units • Data transmission to computer units • Feedback loops in guidance systems • Maintenance checks for sensor units • Troubleshooting common sensor failures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a sensor unit initiates the guidance process.
(b) Describe the role of reference units in maintaining system accuracy.
(c) Identify one potential failure mode in a guidance system computer unit.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts learned in this lesson will be applied in the study of amplifier, controller, actuator and feedback unit of guidance system in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Amplifier, controller, actuator and feedback unit of guidance system.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min